



From Propositions to Arguments

LOGICAL REASONING

LECTURE 2

What is a Proposition?

- ▶ A **proposition** is something that may be **asserted or denied**. It expresses a claim that something **is** or **is not** the case.
- ▶ Because a proposition makes a claim about reality, it must have a **truth value**.
- ▶ Every proposition is therefore either:
 - **True**
 - **False**
- ▶ Even if we do not know whether it is true or false, the proposition still has a truth value.
- ▶ **Example**
Water boils at 100°C.
This proposition expresses a claim about the world that can be **true or false**.

Truth Value of Propositions

- ▶ A key feature of propositions is that they possess a **truth value**.
- ▶ Truth value refers to the logical status of a proposition as being **true or false**.
- ▶ Importantly, we do not need to know the truth of a proposition for it to have a truth value.
- ▶ **Example**
"There is intelligent life on another planet."
 - ▶ We may not know whether this claim is true or false, but the proposition **still has a truth value**.

Sentences that are NOT Propositions

- ▶ Not every sentence expresses a proposition.
- ▶ Only sentences that **assert something capable of being true or false** count as propositions.
- ▶ The following types of sentences do **not express propositions**:
 - Questions
 - Commands
 - Suggestions
 - Proposals
 - Exclamations

Types of Propositions

- ▶ In logic we study two main kinds of propositions:

1. **Simple Propositions**

2. **Compound Propositions**

- ▶ **Simple Proposition**

A **simple proposition** expresses a single claim and does not contain another proposition within it.

- ▶ Example:

Socrates is a man.

Examples of Simple Propositions

- ▶ Examples of simple propositions include:
 - The Earth revolves around the Sun.
Ali passed the examination.
Snow is white.
The library is open today.
- ▶ Each of these expresses **a single independent claim about reality.**

Compound Propositions

- ▶ A **compound proposition** contains two or more propositions connected together.
- ▶ These propositions are joined using **logical connectives**.
- ▶ Common forms of compound propositions include:
 - Negation
 - Conjunction
 - Disjunction
 - Conditional

Negation

- ▶ A **negation** denies a proposition.
- ▶ Negation is usually expressed using the word **not**.
- ▶ **Example**
Original proposition:
Socrates is wise.
- ▶ Negation
Socrates is **not** wise.
- ▶ The negation reverses the truth value of the original claim.

Conjunction

- ▶ A **conjunction** combines two propositions using the word **and**.
- ▶ Each component proposition is called a **conjunct**.
- ▶ **Example**
Smith is neat **and** Smith is polite.
- ▶ This compound proposition contains two claims:
 - Smith is neat
 - Smith is polite
- ▶ The conjunction is true only when **both propositions are true**.

Disjunction

- ▶ A **disjunction** connects two propositions using the word **or**.
- ▶ **Example**
The prisoner has a red hat **or** the prisoner has a white hat.
- ▶ Another example:
Ali will study tonight **or** he will fail the exam.
- ▶ Disjunction presents **alternative possibilities**.

Conditional Propositions

- ▶ A **conditional proposition** has the structure **if-then**.
- ▶ It expresses a relationship between two propositions.
- ▶ **Example**
If the train is late, then we will miss our appointment.
- ▶ This contains two parts:
 - ▶ Antecedent
If the train is late
 - ▶ Consequent
Then we will miss our appointment

From Propositions to Arguments

- ▶ Single propositions by themselves do not constitute reasoning.
- ▶ Reasoning begins when **multiple propositions are connected in such a way that one provides support for another.**
- ▶ This structure is called an **argument**

What is an Argument?

- ▶ An **argument** is a group of statements or propositions in which:
 - one or more statements provide **reasons**
 - another statement is supported as the **conclusion**

The Core Structure of an Argument

- ▶ Every argument has three essential elements:
- ▶ **1. Premises:** Statements that provide reasons or evidence
- ▶ **2. Conclusion:** The statement that is supported
- ▶ **3. Inference:** The reasoning process that connects premises to conclusion
- ▶ Premises → Inference → Conclusion

Example of an Argument

- ▶ Premise
All crimes are violations of the law.
- ▶ Premise
Theft is a crime.
- ▶ Conclusion
Therefore theft is a violation of the law.
- ▶ The premises provide **logical support for the conclusion.**

What is a Premise?

- ▶ A premise is a statement that provides a reason for accepting a conclusion.
- ▶ Premises may take different forms:
General rule, Observation, Fact, Assumption
- ▶ Example:
He must be tired because he worked all night.
Premise: He worked all night
Conclusion: He is tired

Premise Indicators

- ▶ Certain words signal that a statement functions as a **premise**.
- ▶ Common premise indicators include:
since, because, for, given that, seeing that, inasmuch as
for the reason that, owing to the fact that
- ▶ **Example**
 - Ali must be at home **because** his car is outside.
 - Premise
His car is outside.
 - Conclusion
Ali must be at home.

What is a Conclusion?

- ▶ The conclusion is the statement that the argument is trying to establish or prove.
- ▶ It is not just another sentence—it is the main point of the argument.
- ▶ Example:

All humans are mortal.

Socrates is human.

Therefore, Socrates is mortal. (Conclusion)

Conclusion Indicators

- ▶ Certain words indicate that a statement functions as a **conclusion**.
- ▶ Common conclusion indicators include:
Therefore, thus, hence, so, accordingly, consequently
it follows that, we may conclude, we may infer
- ▶ **Example**
- ▶ The road is wet.
Therefore it rained last night.

What is Inference?

- ▶ Inference is the process of reasoning that connects premises to a conclusion.
- ▶ It is not always explicitly stated—it is something the mind performs.
- ▶ Example:
If the lights are on, someone is home.
The lights are on.
Therefore, someone is home.
- ▶ Here, inference is the logical movement from premises to conclusion.

Argument vs Non-Argument

- ▶ Not every group of statements forms an argument.
- ▶ An argument must involve a claim of support.
- ▶ Non-Argument (Description):
The room is large.
It has two windows.
- ▶ No reasoning—just information.
- ▶ Argument:
The room has two windows.
Therefore, it receives a lot of light.
- ▶ Now there is reasoning.